

Arizona State University
Department of Economics
Ph.D. Qualifying Exam

Macroeconomic Theory

June 4, 2020

Instructions: Read the questions carefully, show all your work, and explain all your answers. Indicate carefully if you introduce notation, or make assumptions when answering a question. Please make sure to allocate time to ALL three problems.

Good luck!

1. Consider the following optimal growth model with endogenous labor supply. Households value consumption (c) and leisure (l) in each period. In addition, each household also values the aggregate per-person leisure of *all* households in the economy, $\bar{l}$ (*public leisure*). Specifically, each household has preferences given by

$$E_0 \sum_{t=0}^{\infty} \beta^t [u(c_t) + \phi(l_t, \bar{l}_t)] , \quad (1)$$

where $u(\cdot)$ is strictly increasing, strictly concave and differentiable, $\phi(\cdot, \cdot)$ is strictly increasing in both arguments, strictly concave, and differentiable, and $\beta \in (0, 1)$.

Each household member is endowed with one unit of time that he/she devotes to leisure or to work (h) at date t . Each household consists of one member and there are N identical households in the economy. Population is constant.

The social planner maximizes (1) above subject to

$$\begin{aligned} Nc_t + K_{t+1} &\leq f(K_t, Nh_t) + (1 - \delta)K_t, \\ c_t, K_{t+1}, h_t &\geq 0, \quad l_t + h_t \leq 1 \end{aligned}$$

and $K_0 > 0$, where K stands for the aggregate capital stock, which depreciates at the rate $\delta \in (0, 1]$. The production function $f(\cdot)$ is strictly increasing in both capital and labor, differentiable, and displays diminishing marginal products under constant returns to scale. It also satisfies standard Inada conditions.

- (a) Formulate the planner's problem in the recursive language. Specify carefully the relevant state variables for the planner and the corresponding decision rules.
- (b) Assume that the value function for the planner is differentiable. Derive the first-order conditions and the Envelope condition.
- (c) Show that the capital-labor ratio in steady state is unique and equal to the value in the standard model with no valuation of public leisure. Explain.
- (d) Suppose further that the production function is Cobb-Douglas, the depreciation rate equals 1, and the period utility function is given by

$$u(c_t) + \phi(l_t, \bar{l}_t) = \log(c_t) + b_1 \log(l_t) + b_2 \log(\bar{l}_t),$$

with $b_1, b_2, \gamma > 0$. Solve for the hours worked in steady state in closed form.

- (e) Economist A argues that in a competitive equilibrium for this economy, individuals work “too much”, and that such equilibrium is inefficient. Is she right?

2. This problem asks you to analyze a recursive competitive equilibrium in a McCall search model with many workers and an asset market. There is a continuum of workers of measure one. Each worker maximizes $E \sum_{t=0}^{\infty} \beta^t u(c_t)$, where $c_t \geq 0$ is consumption in period t , u is strictly increasing and strictly concave, and $\beta \in (0, 1)$.

A worker can be either employed or unemployed, and can smooth consumption using a one-period bond traded on a competitive asset market. In the beginning of each period, workers decide how much to save/borrow (subject to a borrowing limit $\underline{a}$). Then each unemployed worker receives a wage offer drawn from a distribution with c.d.f. $F(w)$ and p.d.f. $f(w)$ on $[0, \bar{w}]$. The wage offers are i.i.d. across workers and over time. The unemployed worker decides whether accept or reject the offer. If he rejects, he receives an (exogenous) unemployment benefit b today, and enters the next period as unemployed. If he accepts, he is employed next period at wage w .

An employed worker can lose his job with probability $\rho \in (0, 1)$ at the end of every period. In that case, he becomes unemployed starting from next period (he still receives his wage this period).

- (a) Define a *stationary* recursive competitive equilibrium. You can assume that an unemployed worker's optimal decision rule regarding which wage offers to accept has a reservation-wage form.
- (b) Show that if an employed worker consumes more than an unemployed worker with the same asset level, then the reservation wage is strictly increasing in the asset level.

3. Consider a two-period OLG economy. Time is discrete and continues forever, indexed by t . Individuals live for two periods, labeled “youth” (y) and “old age” (o). The number of young individuals at time t is denoted by N_t^y and that of the old N_t^o . Total population is $N_t = N_t^y + N_t^o$. Young population grows over time according to $N_{t+1}^y = (1 + n)N_t^y$.

An individual’s preferences are described by

$$U(c_t^y, c_{t+1}^o) = \log c_t^y + \beta \log c_{t+1}^o,$$

where $\beta > 0$. The budget constraints in young and old age are

$$\begin{aligned} c_t^y + s_t &= w_t - d_t, \\ c_{t+1}^o &= (1 + r_{t+1})s_t + p_{t+1}, \end{aligned}$$

where s_t is savings, w_t is the wage, d_t is contribution to Social Security, r_{t+1} is the interest rate between period t and $t + 1$, and p_{t+1} is the pension received in old age.

Output Y_t is produced according to a Cobb-Douglas production function $Y_t = K_t^\alpha L_t^{1-\alpha}$, where K_t and L_t denote capital and labor. The depreciation rate of capital is $\delta = 1$.

The government has balanced budget period by period (total contributions equal total pensions).

- (a) Derive an individual’s first-order conditions. Define the competitive equilibrium.
- (b) Assume that Social Security contributions d_t are set according to $d_t = \tau w_t$, where $\tau \in (0, 1)$ is a proportional tax rate. Find the pension amount p_t such that the government budget constraint is balanced period by period. (Hint: Note that $N_t^o = N_{t-1}^y$.)
- (c) Obtain the *steady-state* capital-labor ratio. (Recall that $\delta = 1$.)
- (d) How does the capital-labor ratio changes if one increases the contribution rate τ ? Explain your result intuitively.